

Submission of Self-Practice Evidence for Programming Part 2

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1 Problems

1.1 Problem 1: Remove Duplicates from Sorted Array

- Link: [Leetcode - Remove Duplicates from Sorted Array](#)
- Level: Easy

1.1.1 Description

- The problem told us to return a new array that has removed the duplicate elements.
- I decided to use the `<vector>` library to have a flexible array, along with the `<algorithm>` library to sort the array.
- My idea is...
 1. Let the user prompt the size and elements (in the `main()` function).
 2. Sort the array using `std::sort()`.
 3. Check if the next number is equal to the current number.

```
#include <vector>
#include <algorithm>

using namespace std;

int removeDuplicates(vector<int>& nums)
{
    int s = 0;

    sort(nums.begin(), nums.end());

    int c = 0;
```

```

while (c < nums.size()) {
    if (nums[c] == nums[c + 1])
    {
        s = c+1;
        nums.erase(nums.begin() + s);
        c--;
    }

    c++;
}

return nums.size();
}

```

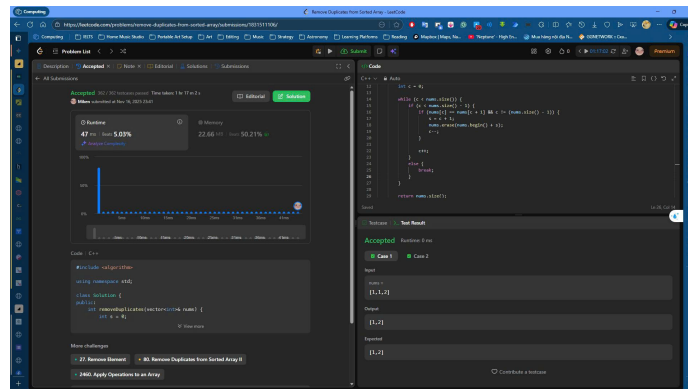


Figure 1: Leetcode - Remove Duplicates from Sorted Array

1.1.2 Test cases

1. Test case 1: [1, 2, 2, 4, 4] - Output: [1, 2, 4]
2. Test case 2: [3, 3, 4, 1, 2] - Output: [1, 2, 3, 4]
3. Test case 3: [1, 4, 5, 5, 6] - Output: [1, 4, 5, 6]